

Ans: A

- 13 Two identical vessels contain two different gases X and Y with molecular mass m_x and m_y respectively. The thermodynamic temperature of gas X is twice that of gas Y and the root-mean-square speed of the molecules in X is three times that of the molecules in Y. What is the ratio of $\frac{m_x}{m_y}$?

A 2 : 3

B 2 : 9

C 9 : 2

D 3 : 2

Ans: B

Using $KE = \frac{3}{2} kT \rightarrow mv_{rms}^2 \propto T$

$m \propto T/v_{rms}^2$

$$\frac{m_x}{m_y} = \left(\frac{2}{1}\right) \left(\frac{1}{3}\right)^2 = \frac{2}{9}$$

- 14 An ideal monatomic gas has 1000 J of heat added to it and it does 500 J of work: its